Analysis and Forecasting of Weekly Electricity Consumption Using ARIMA Model

Evidence from PJM East Electricity Load Data

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Abstract

This report analyzes and forecasts weekly electricity consumption in the PJM East region using an ARIMA-based time series approach. The original hourly electricity load data from `PJME_hourly.csv` are cleaned, checked for missing and duplicated timestamps, and aggregated into weekly electricity consumption by summing hourly values. The final dataset contains 864 weekly observations from 13 January 2002 to 29 July 2018.

The weekly series is split chronologically into training, validation, and test sets. Naive and Seasonal Naive models are used as benchmarks, while several ARIMA specifications are compared using validation performance. Since the ADF test suggests stationarity, the differencing order is set to $d = 0$. The selected ARIMA(3,0,2) model achieves a final test MAPE of 4.86%, corresponding to an average forecast accuracy of approximately 95.14%. A rolling one-step forecast gives a slightly lower MAPE of 4.45%. The results indicate that ARIMA improves forecasting accuracy compared with simple baseline methods, although annual seasonality and the absence of external variables remain important limitations.

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1 Introduction

Electricity consumption forecasting is important for energy planning and power system operation. Reliable forecasts help utilities anticipate future demand, allocate resources, and maintain the stability of electricity supply. Since electricity consumption changes over time and often follows repeated patterns, time series forecasting provides a useful framework for analyzing and predicting future demand.

This report focuses on forecasting electricity consumption in the PJM East region using the Autoregressive Integrated Moving Average (ARIMA) model. The data are taken from the *Hourly Energy Consumption* Kaggle dataset Mulla (2018), specifically the `PJME_hourly.csv` file. The original dataset contains hourly electricity load observations, measured by the variable `PJME_MW`. Since the raw data are recorded at an hourly frequency, they must be cleaned and transformed before being used for weekly electricity consumption forecasting.

The main research question of this report is:

> Can an ARIMA model accurately forecast weekly electricity consumption in the PJM East region using historical electricity load data?

To answer this question, the report follows a structured time series forecasting procedure. First, the raw hourly data are cleaned by checking duplicated timestamps, missing hourly observations, and incomplete weeks. The hourly load values are then aggregated into weekly electricity consumption by summing all hourly observations within each week. This transformation reduces high-frequency fluctuations while preserving meaningful weekly demand patterns.

After constructing the final weekly series, exploratory data analysis is used to examine the overall movement, distribution, and variability of electricity consumption. The dataset is then split chronologically into training, validation, and test sets. Naive and Seasonal Naive models are used as baseline forecasts, while several ARIMA specifications are estimated and compared using validation performance. The final selected model is evaluated on the test set using MAE, RMSE, and MAPE.

This report has three main objectives. First, it constructs a reproducible weekly forecasting dataset from the original hourly electricity load data. Second, it evaluates whether an ARIMA model can improve forecasting accuracy compared with simple baseline methods. Third, it discusses the strengths and limitations of using a univariate ARIMA model for weekly electricity consumption forecasting.

2 Literature Review

Electricity consumption forecasting is important for power system operation, energy planning, and resource management. Since electricity demand must be balanced with supply, inaccurate forecasts may lead to inefficient generation planning, higher operating costs, or reliability problems. Previous studies emphasize that accurate electricity demand forecasting supports both utility companies and policy makers in planning future energy requirements Tarmanini et al. (2023); Izudin et al. (2021).

Time series models are commonly used in electricity forecasting because they rely on historical observations to predict future values. Among these methods, ARIMA is a widely used statistical forecasting model because it captures temporal dependence through autoregressive, differencing, and moving average components. Chodakowska et al. Chodakowska et al. (2021) examine ARIMA models in electrical load forecasting and highlight that load data often contain noise and irregular movements. Tarmanini et al. Tarmanini et al. (2023) also compare ARIMA with artificial neural network approaches for short-term load forecasting, showing that ARIMA can serve as a useful statistical benchmark.

Previous studies also suggest that ARIMA has limitations when electricity consumption contains nonlinear patterns or is affected by external factors. Izudin et al. Izudin et al. (2021) discuss hybrid ARIMA-ANN models for electricity consumption forecasting, where ARIMA is used to capture linear time series behavior and ANN is used to capture nonlinear components. This supports the use of ARIMA as an interpretable baseline model, while also recognizing that more complex models may be needed when nonlinear effects or external variables are important.

Seasonality is another important issue in electricity forecasting. Electricity demand may vary by hour, day, week, and season. For weekly data, annual seasonality may appear around lag 52, representing the same week in the previous year. Hyndman and Athanasopoulos Hyndman and Athanasopoulos (2021) describe Naive and Seasonal Naive methods as simple benchmark forecasting approaches. These methods are useful because they provide reference points for evaluating whether a more advanced model improves forecasting accuracy.

The dataset used in this report is the *Hourly Energy Consumption* dataset from Kaggle, specifically the `PJME_hourly.csv` file Mulla (2018). Unlike studies that forecast hourly load directly, this report aggregates the original hourly data into weekly electricity consumption. This transformation reduces high-frequency fluctuations and allows the analysis to focus on weekly consumption patterns. Based on the literature, this report compares the selected ARIMA model with Naive and Seasonal Naive baselines to evaluate whether ARIMA provides meaningful forecasting improvement.

3 Data Description and Preprocessing

3.1 Data Source

This report uses the *Hourly Energy Consumption* dataset from Kaggle, specifically the `PJME_hourly.csv` file Mulla (2018). The dataset contains hourly electricity load observations for the PJM East region. The main variable is `PJME_MW`, which measures electricity load in megawatts (MW).

Because the raw data are recorded hourly, preprocessing is required before constructing the weekly forecasting series. The datetime column is converted into a datetime format, set as the index, and sorted chronologically. Duplicated timestamps and missing hourly observations are checked because time series forecasting requires a consistent and correctly ordered time index.

3.2 Missing Values, Outliers, and Weekly Aggregation

After reindexing the data to a complete hourly frequency, missing values in PJME_MW are identified and treated using interpolation. This is appropriate because electricity load is a continuous time series and nearby hourly observations provide useful information for estimating small gaps.

Extreme values are inspected but not removed automatically. In electricity load data, unusually high or low observations may reflect real peak-demand or low-demand periods rather than data errors. Therefore, observations are retained unless they are clearly related to data quality problems such as duplicates or missing timestamps.

The original variable PJME_MW measures hourly electricity load. Since the objective of this report is to forecast electricity consumption, hourly values are aggregated into weekly total consumption:

$$Weekly\_Electricity\_Consumption\_MWh_t = \sum_{h \in t} PJME\_MW_h,$$

where t represents a given week and h represents the hourly observations within that week. Because the data are recorded hourly, summing hourly MW values gives an approximation of weekly electricity consumption in megawatt-hours (MWh).

Weekly aggregation is chosen because it provides a balance between reducing high-frequency hourly and daily fluctuations while retaining enough observations for time series modeling. Hourly data contain strong intraday variation, daily data may still contain day-of-week effects, and monthly aggregation would produce a much smaller sample. Therefore, weekly aggregation is suitable for short-term and medium-term electricity consumption forecasting.

3.3 Final Forecasting Dataset

After aggregation, the number of hourly observations in each week is checked. A complete week should contain 168 hourly observations. Incomplete weeks are removed to avoid artificially low weekly totals. The final dataset contains 864 weekly observations from 13 January 2002 to 29 July 2018. The target variable is Weekly_Electricity_Consumption_MWh.

Table 1: Summary of the Final Forecasting Dataset

Item	Description
Original dataset	Hourly Energy Consumption, PJME_hourly.csv
Original region	PJM East
Original variable	PJME_MW
Original frequency	Hourly
Forecasting variable	Weekly_Electricity_Consumption_MWh
Final frequency	Weekly, ending on Sunday
Aggregation method	Sum of hourly load values within each week
Final sample period	13 January 2002 to 29 July 2018
Number of observations	864 weekly observations
Final unit	MWh

The final weekly dataset is used for exploratory data analysis, baseline forecasting, ARIMA model selection, and final forecast evaluation. Model identification and selection are conducted only after the data are split chronologically into training, validation, and test sets to avoid data leakage.

4 Exploratory Data Analysis

4.1 Overall Pattern of Weekly Electricity Consumption

Figure 1 presents the weekly electricity consumption series constructed from the original hourly PJM East load data. Each observation represents total electricity consumption within one week, measured in MWh.

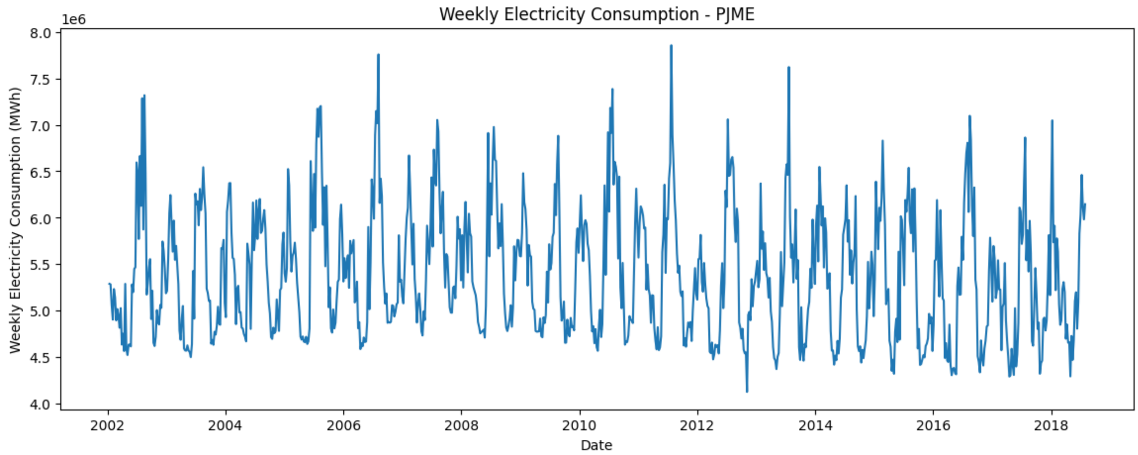

Figure 1: Weekly electricity consumption over time.

The figure shows that weekly electricity consumption fluctuates considerably over the sample period. Most weekly values are between approximately 4 million and 7 million MWh, while several peak-demand weeks reach higher levels. These peaks are important because they may reflect genuine periods of high electricity demand rather than data errors. Therefore, they are retained in the dataset.

The series does not follow a simple linear upward or downward trend over the entire period. Instead, it shows repeated increases and decreases over time. This suggests that electricity consumption may contain seasonal or cyclical patterns. Since electricity demand is often affected by weather, calendar effects, and economic activity, such repeated movements are expected in electricity load data.

This initial plot also supports the choice of weekly aggregation. Compared with hourly or daily data, the weekly series is smoother and easier to interpret, while still preserving meaningful variation in electricity consumption.

4.2 Rolling Mean and Rolling Standard Deviation

Figure 2 shows the weekly electricity consumption series together with the 4-week rolling mean and 4-week rolling standard deviation. The rolling mean smooths short-term fluctuations and helps identify the underlying movement of the series. The rolling standard deviation provides information about changes in short-term variability.

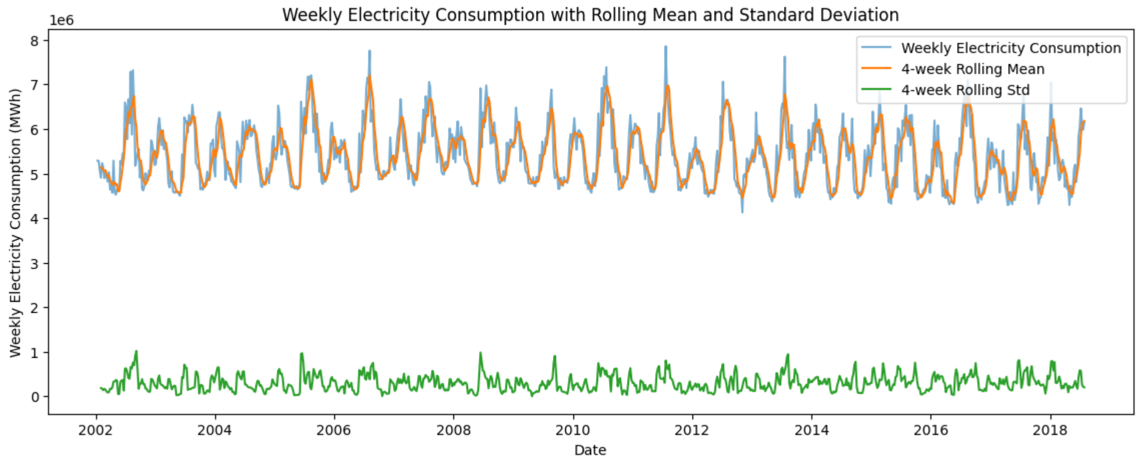

Figure 2: Weekly electricity consumption with 4-week rolling mean and rolling standard deviation.

The rolling mean follows the main movements of the weekly consumption series and highlights repeated periods of rising and falling electricity demand. This suggests that the weekly series contains systematic time-dependent behavior rather than random fluctuations only. The rolling mean also shows that high-consumption periods tend to occur in clusters, which is consistent with seasonal demand patterns.

The rolling standard deviation changes over time, indicating that the variability of weekly electricity consumption is not perfectly constant. Some periods show stronger short-term fluctuations than others. This is relevant for forecasting because models such as ARIMA may perform better during stable periods and worse during periods with sudden demand changes.

However, this figure is used only for descriptive analysis. It provides an initial visual understanding of the data, but it is not used directly to select the ARIMA order. Formal stationarity testing and autocorrelation analysis are conducted later using the training set.

4.3 Distribution of Weekly Electricity Consumption

Figure 3 presents the distribution of weekly electricity consumption. This histogram shows how frequently different levels of weekly consumption occur in the final dataset.

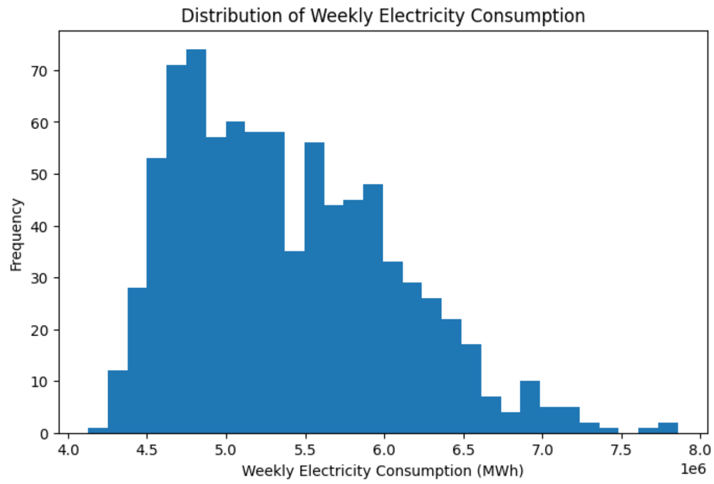

Figure 3: Distribution of weekly electricity consumption.

The distribution is not perfectly symmetric. Most weekly electricity consumption values are concentrated between approximately 4.5 million and 6.0 million MWh. The histogram also shows a right tail, with some weeks having consumption above 7 million MWh. These high-consumption weeks represent peak-demand observations.

The presence of a right tail is important for interpreting forecast errors. Metrics such as RMSE are sensitive to large errors, so weeks with unusually high consumption may increase RMSE if the model fails to forecast them accurately. For this reason, this report evaluates forecasting performance using MAE, RMSE, and MAPE together rather than relying on a single metric.

The histogram also supports the earlier decision not to remove extreme values automatically. Since high weekly consumption values may reflect real peak-demand periods, removing them could eliminate important information from the dataset. Instead, these observations are retained and treated as part of the electricity consumption behavior that the forecasting model should learn.

Overall, the exploratory analysis shows that the weekly electricity consumption series contains substantial variation, repeated temporal movements, and some high-demand observations. These patterns motivate the use of time series forecasting methods and justify the later analysis of stationarity, autocorrelation, baseline forecasts, and ARIMA model selection.

It is important to note that the figures in this section are used only for descriptive purposes. Formal model identification and model selection are conducted later using the training and validation sets in order to avoid data leakage.

5 Methodology

5.1 Train-Validation-Test Split

Since the data are time series observations, the dataset is split chronologically rather than randomly. Random splitting is not appropriate for time series forecasting because it would break the temporal order of the data and may allow future information to influence the model estimation process. Therefore, this report preserves the time order of the weekly electricity consumption series.

The final weekly dataset is divided into three parts: training set, validation set, and test set. The training set is used for stationarity testing, autocorrelation analysis, and model estimation. The validation set is used to compare candidate ARIMA models and select the preferred specification. The test set is kept unseen during model selection and is used only for final out-of-sample evaluation.

In this report, the validation set contains 52 weekly observations, corresponding to approximately one year of data. The test set contains 13 weekly observations, corresponding to approximately one quarter. The remaining earlier observations are used as the training set. Table 2 summarizes the chronological split.

Table 2: Chronological Train-Validation-Test Split

Subset	Period	Number of Observations	Purpose
Training set	13 Jan 2002 – 30 Apr 2017	799	For stationarity testing, autocorrelation analysis, and model estimation.
Validation set	07 May 2017 – 29 Apr 2018	52	For comparing candidate ARIMA specifications and selecting the preferred model.
Test set	06 May 2018 – 29 Jul 2018	13	Reserved for final out-of-sample forecast evaluation.

Figure 4 illustrates the chronological split of the weekly electricity consumption series.

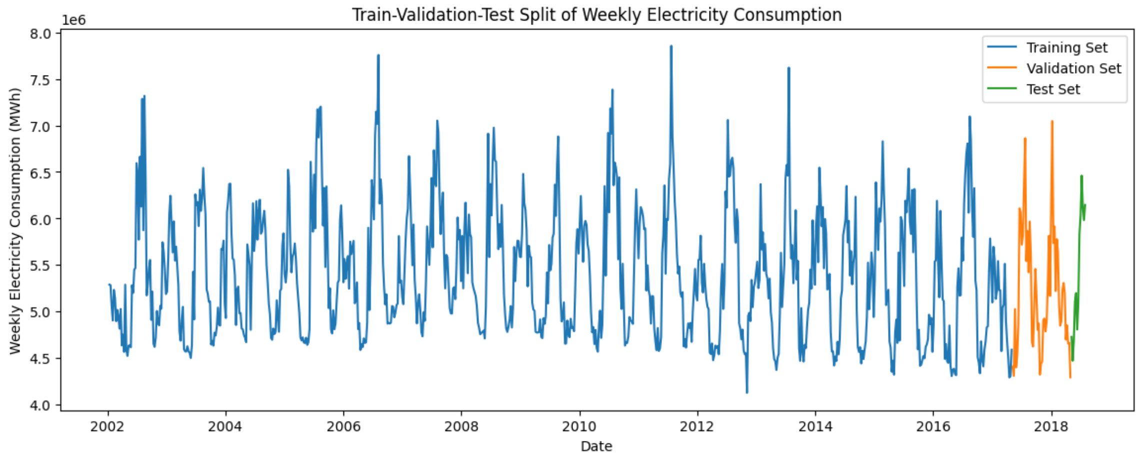

Figure 4: Chronological train-validation-test split of weekly electricity consumption.

This split ensures that the test set is not used during model identification or model selection. As a result, the final evaluation reflects the model's ability to forecast unseen future observations.

5.2 Baselines for Forecasting Model Evaluation

As a starting point for fitting ARIMA models, this report uses simple baseline forecasting models as comparison methods. Naive and Seasonal Naive forecasts are commonly used as benchmark methods in time series forecasting Hyndman and Athanasopoulos (2021).

First, we employ the Naive forecasting approach. The Naive forecast postulates that future values would be identical to the latest observed value. Thus, if y_t is the latest observed value, then the h-step ahead Naive forecast would be calculated as follows:

$$\hat{y}_{t+h} = y_t.$$

Second, we consider the Seasonal Naive forecast approach. In order to account for the weekly pattern of our final data, we define a seasonal lag of 52, where the number 52 stands for the same week a year earlier. Under the Seasonal Naive forecasting approach, the future values of weekly electricity consumption will be estimated by the respective values from the previous year:

$$\hat{y}_{t+h} = y_{t+h-52}.$$

The Naive baseline would help us evaluate whether the proposed ARIMA model is any better than persistence-based forecasting. The Seasonal Naive baseline serves as an enhanced benchmark, as it takes into account the potential annual seasonality of weekly electricity consumption.

5.3 Stationarity Testing

ARIMA modeling requires the time series to be stationary or transformed into a stationary series through differencing. A stationary series has statistical properties such as mean and variance that are relatively stable over time. To examine stationarity, this report applies the Augmented Dickey-Fuller (ADF) test to the training set.

The null hypothesis of the ADF test is that the series has a unit root, meaning that it is non-stationary. The alternative hypothesis is that the series is stationary. If the p-value is below the 5% significance level, the null hypothesis is rejected.

The ADF test result for the weekly training series is summarized in Table 3.

Table 3: ADF Test Result for the Weekly Training Series

Statistic	Value
ADF statistic	-12.3704
p-value	5.30×10^{-23}
1% critical value	-3.4387
5% critical value	-2.8652
10% critical value	-2.5687

Since the p-value is far below 0.05, the null hypothesis of a unit root is rejected. Therefore, the weekly training series is treated as stationary, and the differencing order is set to $d = 0$.

5.4 ARIMA Model Identification

After confirming stationarity, the next step is to identify candidate ARIMA models. An ARIMA model is denoted as ARIMA(p, d, q), where p is the autoregressive order, d is the differencing order, and q is the moving average order. Since the ADF test indicates that the training series is stationary, this report focuses on ARIMA$(p, 0, q)$ models.

The autocorrelation function (ACF) and partial autocorrelation function (PACF) are used to examine the dependence structure of the training series. The ACF shows the correlation between the series and its lagged values, while the PACF shows the partial correlation after controlling for shorter lags. These plots provide initial guidance for selecting candidate values of p and q.

Figure 5 presents the ACF and PACF plots of the weekly training series.

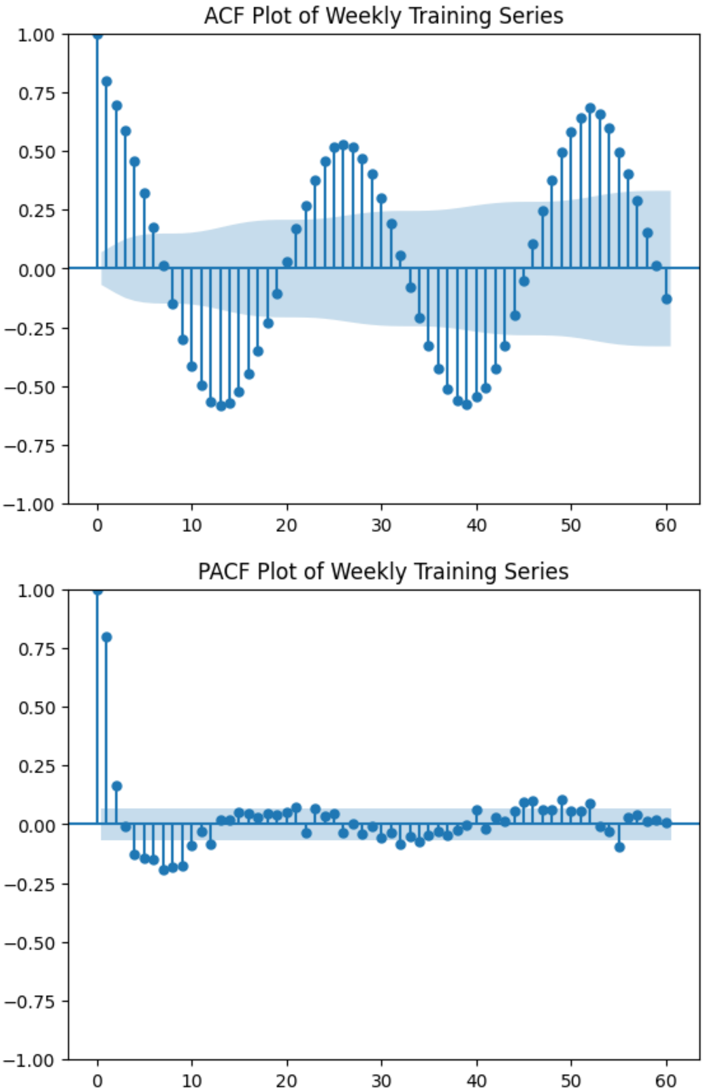

Figure 5: ACF and PACF plots of the weekly training series.

The ACF plot shows strong autocorrelation at short lags and a repeated pattern around longer lags, suggesting that weekly electricity consumption contains time-dependent behavior and possible annual seasonality around lag 52. The PACF plot shows strong partial autocorrelation at the first lag and weaker values at later lags. These patterns motivate the use of ARIMA models with several candidate autoregressive and moving average orders.

However, the final ARIMA specification is not selected only by visual inspection of the ACF and PACF plots. Instead, several candidate models are estimated and compared using validation forecasting performance.

5.5 Model Selection Strategy

Candidate ARIMA models are fitted on the training set and evaluated on the validation set. The validation set is used because it allows different model specifications to be compared without using the test set. This prevents the test set from influencing model selection.

For each candidate ARIMA$(p, 0, q)$ model, the fitted model produces forecasts for the validation period. The validation forecasts are then compared with the actual validation values using forecasting error metrics. The candidate models are compared using MAE, RMSE, and MAPE. The preferred model is selected based on validation performance and model simplicity, while residual diagnostics are examined after model selection to evaluate remaining autocorrelation.

After the final ARIMA specification is selected, the model is refitted using the combined training and validation sets. The final fitted model is then used to forecast the test set. The test set remains unseen until this final evaluation step.

5.6 Forecast Evaluation Metrics

Forecasting performance is evaluated using three standard error metrics: mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE).

The MAE is defined as:

$$MAE = \frac{1}{n}\sum_{t=1}^{n} |y_t - \hat{y}_t|,$$

where y_t is the actual value, $\hat{y}_t$ is the forecast value, and n is the number of forecasted observations.

The RMSE is defined as:

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}.$$

RMSE gives more weight to large forecast errors because the errors are squared before averaging. Therefore, RMSE is useful for identifying whether the model performs poorly during periods with unusually large forecast errors.

The MAPE is defined as:

$$MAPE = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right|.$$

MAPE is useful because it expresses forecast errors as percentages, making it easier to interpret forecasting accuracy relative to the scale of the data. Since weekly electricity consumption is measured in millions of MWh, MAE and RMSE can appear large in absolute terms. Therefore, this report interprets MAE and RMSE together with MAPE and baseline comparisons.

Forecast accuracy is also reported as:

$$Forecast\ Accuracy = 100 - MAPE.$$

This measure is used only as a descriptive interpretation of the average percentage error.

5.7 Multi-step and Rolling One-step Forecasting

The main final test evaluation in this report uses a multi-step forecast. In the multi-step setting, the final ARIMA model is fitted using the combined training and validation sets, and then forecasts the entire test period without updating the model with actual test observations. This provides a strict out-of-sample evaluation because the model does not use any information from the test period during forecasting.

In addition to the main multi-step forecast, this report also conducts a rolling one-step forecast as a supplementary short-term forecasting check. In the rolling approach, the model forecasts one week ahead. After the actual value for that week becomes available, it is added to the historical data before forecasting the next week. This process is repeated throughout the test period.

The rolling one-step forecast does not represent a fixed forecast of the entire test horizon from the initial forecast date. Instead, it represents a real-time short-term forecasting exercise where new weekly observations become available over time. Therefore, the multi-step forecast is used as the main final evaluation, while the rolling one-step forecast is reported as an additional check of short-term forecasting performance.

6 Results

6.1 Baseline Forecasting Results

Before evaluating the ARIMA model, two baseline models are first tested on the validation set. The purpose of this comparison is to provide simple benchmark forecasts and to determine whether the ARIMA model can improve forecasting performance beyond basic forecasting rules.

Table 4 presents the validation performance of the Naive and Seasonal Naive baseline models.

Table 4: Baseline Model Performance on the Validation Set

Model	MAE	RMSE	MAPE (%)	Accuracy (%)
Seasonal Naive Baseline	432,658.69	603,947.01	8.24	91.76
Naive Baseline	696,455.96	907,398.43	12.27	87.73

The results show that the Seasonal Naive baseline performs better than the simple Naive baseline. The Seasonal Naive model achieves a validation MAPE of 8.24%, compared with 12.27% for the Naive baseline. This indicates that using the same week from the previous year provides more useful information than simply using the last observed weekly value. This result also suggests that weekly electricity consumption contains annual seasonal information.

Figure 6 compares the actual validation values with the baseline forecasts.

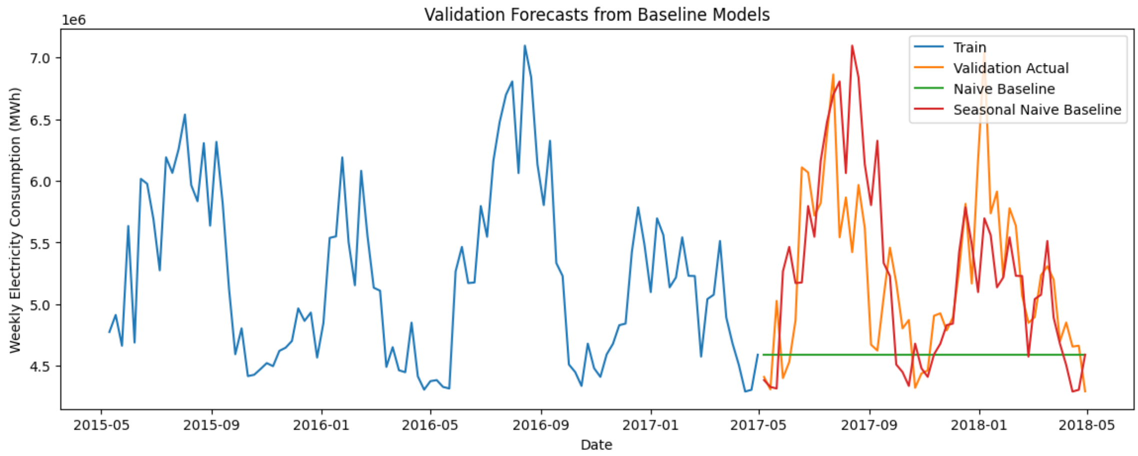

Figure 6: Validation forecasts from the baseline models.

The Naive baseline produces a nearly flat forecast because it uses only the last observed value from the training set. As a result, it cannot capture the fluctuations in the validation period. In contrast, the Seasonal Naive forecast follows the validation series more closely because it uses the same week from the previous year. However, the Seasonal Naive model still produces large forecast errors during some peak-demand and low-demand periods.

6.2 ARIMA Model Selection Results

After evaluating the baseline models, several ARIMA candidate models are estimated using the training set and evaluated on the validation set. Since the ADF test suggests that the weekly training series is stationary, the differencing order is fixed at $d = 0$. Candidate ARIMA$(p, 0, q)$ models are compared using validation MAE, RMSE, and MAPE.

Table 5 reports the main ARIMA candidate models with the best validation performance.

Table 5: ARIMA Candidate Model Selection on the Validation Set

Model	AIC	BIC	MAE	RMSE	MAPE (%)
ARIMA(4,0,3)	22629.43	22671.53	356,944.07	439,977.44	6.93
ARIMA(3,0,2)	22669.65	22702.41	367,409.68	455,106.57	7.23
ARIMA(2,0,2)	22688.37	22716.45	389,314.57	478,645.09	7.67
ARIMA(4,0,1)	22686.91	22719.65	444,846.95	541,492.84	8.81
ARIMA(4,0,0)	22764.97	22793.03	535,452.57	635,705.19	10.34

Based only on validation error, ARIMA(4,0,3) gives the lowest MAPE and RMSE. However, this report selects ARIMA(3,0,2) as the final model because it provides a better balance between forecasting accuracy, model simplicity, and diagnostic performance. The difference in validation MAPE between ARIMA(4,0,3) and ARIMA(3,0,2) is relatively small, while ARIMA(3,0,2) is more parsimonious and easier to interpret. Therefore, ARIMA(3,0,2) is retained as the selected model for final forecasting.

6.3 Validation Comparison between Baseline Models and ARIMA

Table 6 compares the selected ARIMA(3,0,2) model with the two baseline models on the validation set.

Table 6: Validation Comparison between Baseline Models and ARIMA

Model	MAE	RMSE	MAPE (%)	Accuracy (%)
Weekly ARIMA(3,0,2)	367,409.68	455,106.57	7.23	92.77
Seasonal Naive Baseline	432,658.69	603,947.01	8.24	91.76
Naive Baseline	696,455.96	907,398.43	12.27	87.73

The validation results show that the selected ARIMA(3,0,2) model outperforms both baseline models. Compared with the Seasonal Naive baseline, the ARIMA model reduces validation MAPE from 8.24% to 7.23%. The improvement is moderate in percentage-point terms, but the reduction in RMSE is more substantial. This suggests that the ARIMA model captures additional time-series dependence beyond simple annual seasonal repetition.

6.4 Final Test Forecast Results

After selecting ARIMA(3,0,2), the model is refitted using the combined training and validation sets. The final model is then used to forecast the test set. This final evaluation is based on a multi-step forecast, meaning that the entire 13-week test period is forecast without updating the model with actual test observations.

Table 7 compares the final test performance of ARIMA(3,0,2) with the Naive and Seasonal Naive baselines.

Table 7: Final Test Comparison between Baseline Models and ARIMA

Model	MAE	RMSE	MAPE (%)	Accuracy (%)
Weekly ARIMA(3,0,2)	250,006.38	332,071.75	4.86	95.14
Seasonal Naive Baseline	478,081.92	554,567.40	8.74	91.26
Naive Baseline	1,141,057.85	1,313,851.87	19.84	80.16

The test results confirm that ARIMA(3,0,2) provides the best forecasting performance among the three models. The selected ARIMA model achieves a test MAPE of 4.86%, corresponding to an average forecast accuracy of approximately 95.14%. The Seasonal Naive baseline has a higher test MAPE of 8.74%, while the Naive baseline performs much worse with a MAPE of 19.84%.

Although the RMSE value of 332,071.75 MWh may appear large in absolute terms, it should be interpreted relative to the scale of weekly electricity consumption, which is measured in millions of MWh. The relatively low MAPE indicates that the average percentage error is small. Therefore, the model performs well in terms of relative forecasting accuracy.

Figure 7 presents the final test forecast from the selected ARIMA(3,0,2) model.

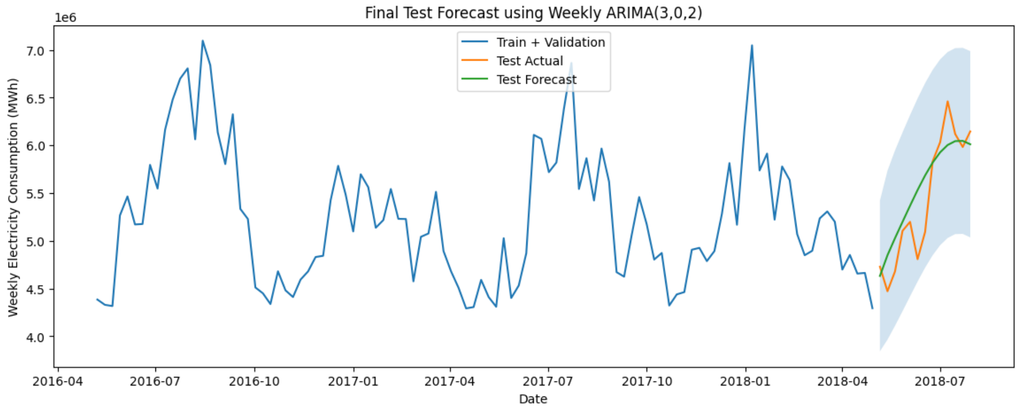

Figure 7: Final test forecast using Weekly ARIMA(3,0,2).

The forecast plot shows that the ARIMA model captures the general upward movement in the test period. However, the forecast line is smoother than the actual test series. This is expected because the multi-step forecast is produced without updating the model with actual test observations. As a result, the model can capture the general direction of consumption but may not fully follow all short-term fluctuations.

6.5 Residual Diagnostics

After selecting ARIMA(3,0,2), residual diagnostics are examined to evaluate whether the model captures the main time-dependent structure in the weekly electricity consumption series. In time series forecasting, residuals should ideally behave approximately like white noise. If significant autocorrelation remains in the residuals, this suggests that the model has not fully captured all dependence in the data.

Figure 8 shows the residuals from the fitted ARIMA(3,0,2) model. The residuals fluctuate around zero, suggesting that the model does not leave a strong visible trend in the errors. However, some large positive and negative residuals remain, indicating that the model does not fully capture all peak-demand and low-demand movements.

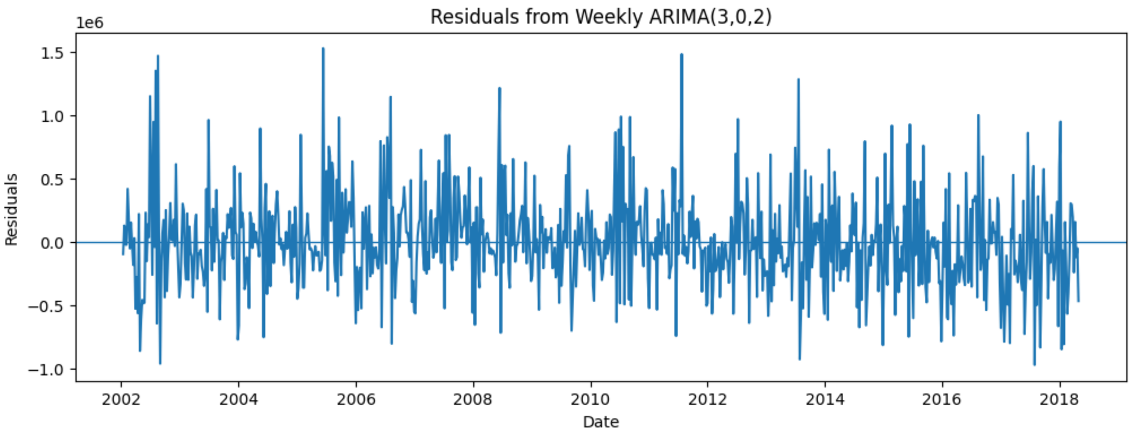

Figure 8: Residuals from Weekly ARIMA(3,0,2).

Figure 9 presents the ACF plot of the ARIMA residuals. Most residual autocorrelations are relatively small, especially at short lags. This suggests that the selected ARIMA model captures much of the short-term dependence in weekly electricity consumption.

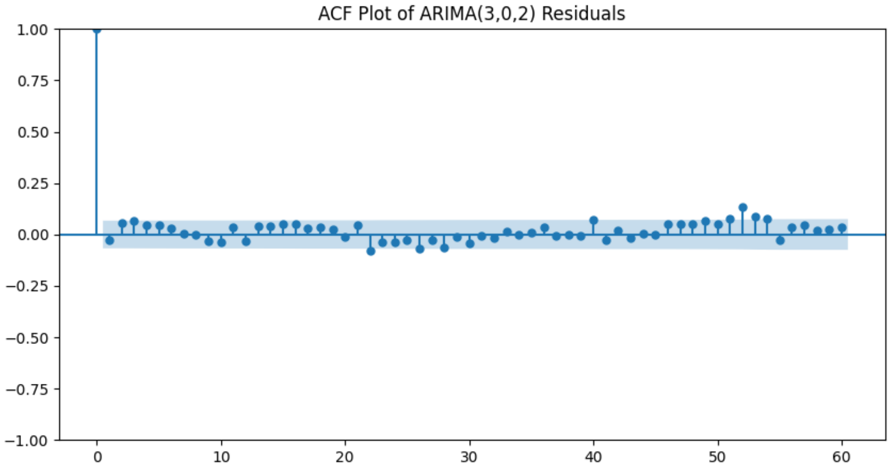

Figure 9: ACF plot of residuals from Weekly ARIMA(3,0,2).

Table 8 reports the Ljung-Box test results for selected residual lags. The selected lags include short-term lags and lag 52, which corresponds approximately to one year in weekly data.

Table 8: Ljung-Box Test for ARIMA(3,0,2) Residuals

Lag	LB Statistic	p-value
7	11.0001	0.1386
13	16.4644	0.2250
26	39.3311	0.0453
52	87.4593	0.0015

The Ljung-Box results show that residual autocorrelation is not statistically significant at lags 7 and 13. This indicates that the model captures most short-term autocorrelation. However, the p-value becomes significant at lag 26 and especially at lag 52. Since lag 52 represents approximately one year in weekly data, this suggests that some annual seasonal dependence remains in the residuals. Therefore, although ARIMA(3,0,2) performs well in forecasting, it does not fully remove annual seasonal autocorrelation.

6.6 Rolling One-Step Forecast Check

In addition to the main multi-step forecast, a rolling one-step forecast is conducted as a supplementary short-term forecasting check. In this approach, the model forecasts one week ahead. After the actual value for that week becomes available, it is added to the historical data before forecasting the next week.

Table 9 compares the main multi-step forecast with the rolling one-step forecast.

Table 9: Comparison between Multi-step and Rolling One-step ARIMA Forecasts

Forecast Method	MAE	RMSE	MAPE (%)	Accuracy (%)
Rolling One-step ARIMA Forecast	235,833.86	287,700.39	4.45	95.55
Multi-step ARIMA Forecast	250,006.38	332,071.75	4.86	95.14

The rolling one-step forecast performs slightly better than the multi-step forecast, with a lower MAPE of 4.45%. This improvement is expected because the rolling approach updates the information set after each observed week. Therefore, it reflects a real-time short-term forecasting situation rather than a fixed 13-week-ahead forecast.

Figure 10 compares the actual test values, the multi-step ARIMA forecast, and the rolling one-step ARIMA forecast.

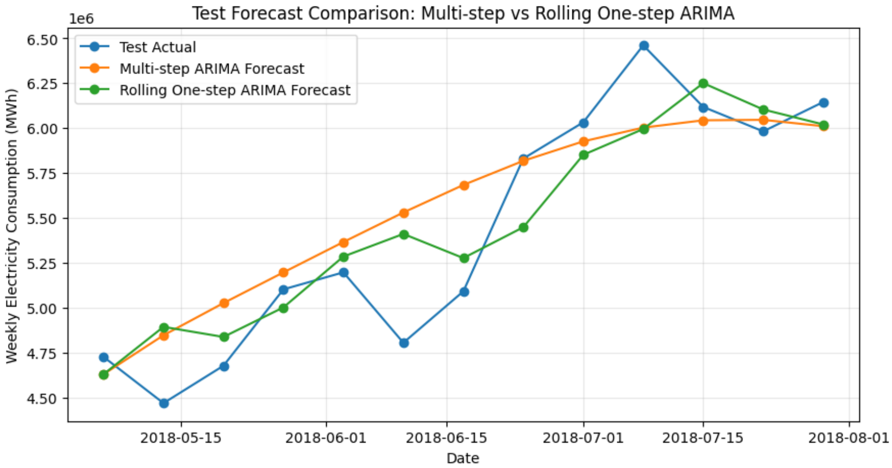

Figure 10: Test forecast comparison between multi-step and rolling one-step ARIMA forecasts.

The figure shows that the rolling one-step forecast follows the actual test values more closely than the multi-step forecast in several weeks. This result suggests that the selected ARIMA model is particularly useful for short-term weekly forecasting when new observations are continuously incorporated. However, the multi-step forecast remains the main final evaluation because it forecasts the entire test period without using actual test observations during the forecasting process.

7 Discussion

The results show that ARIMA(3,0,2) performs better than both Naive and Seasonal Naive baseline models. The Naive baseline performs poorly because it only repeats the last observed weekly value and cannot adjust to changes in demand. The Seasonal Naive baseline performs better because it uses the same week from the previous year, which is consistent with the annual pattern suggested by the ACF plot. However, it still cannot adjust flexibly when the current year differs from the previous year.

Compared with the Seasonal Naive baseline, ARIMA(3,0,2) achieves lower validation and test errors. On the test set, the ARIMA model reduces MAPE from 8.74% for the Seasonal Naive baseline to 4.86%. Although the RMSE value is large in absolute terms, this is mainly because weekly electricity consumption is measured in millions of MWh. Therefore, RMSE should be interpreted together with MAPE and the baseline comparison.

The final multi-step forecast captures the general movement of the test period, but the forecast line is smoother than the actual series. This is expected because the model forecasts the full test horizon without updating with actual test observations. The rolling one-step forecast performs slightly better because the information set is updated after each observed week. Therefore, rolling forecasting is useful as a short-term forecasting check, while the multi-step forecast remains the main final evaluation.

Table 10: SARIMA Validation Results with Seasonal Period 52

Model	Seasonal Order	MAE	RMSE	MAPE (%)
SARIMA(3,0,2)	(1,0,1,52)	415,150.59	516,690.97	8.07
SARIMA(3,0,2)	(1,0,0,52)	422,167.30	525,162.31	8.12
SARIMA(3,0,2)	(0,0,1,52)	466,428.01	564,056.77	8.98

The ACF plot suggests possible annual seasonality around lag 52. Therefore, SARIMA models with seasonal period 52 were also considered. In theory, SARIMA extends ARIMA by adding seasonal autoregressive and moving average components Hyndman and Athanasopoulos (2021). However, Table 10 shows that the tested SARIMA specifications do not improve validation performance. The best SARIMA model has a validation MAPE of 8.07%, while ARIMA(3,0,2) achieves a lower validation MAPE of 7.23%. Therefore, SARIMA is not selected as the final model.

Overall, ARIMA(3,0,2) is suitable for weekly electricity consumption forecasting in this dataset. It outperforms simple benchmark methods and performs well in both multi-step and rolling one-step evaluation. However, residual diagnostics suggest that some annual seasonal autocorrelation remains, and the absence of external variables such as weather and holidays limits the model's ability to explain sudden demand changes.

8 Limitations and Future Research

Although ARIMA(3,0,2) performs well, several limitations remain. First, the model is univariate and uses only past weekly electricity consumption values. In reality, electricity demand is affected by external factors such as temperature, weather conditions, holidays, weekends, and economic activity. Since these variables are not included, the model may not fully capture sudden demand changes.

Second, residual diagnostics indicate that some annual autocorrelation remains around lag 52. Although SARIMA models with seasonal period 52 were tested, they did not improve validation performance compared with ARIMA(3,0,2). Therefore, remaining seasonal dependence should be interpreted as a limitation of the selected non-seasonal ARIMA model.

Third, the test set contains only 13 weekly observations, corresponding to approximately one quarter. A longer test period or rolling-origin evaluation over multiple forecast windows would provide a more complete assessment of model stability. Finally, weekly aggregation reduces high-frequency noise, but it also removes intraday and day-of-week details.

Future research could extend this report by using ARIMAX or SARIMAX models with weather, holiday, and calendar variables Lee et al. (2021); Elamin and Fukushige (2018). It could also compare ARIMA-based models with machine learning methods such as Random Forest, XGBoost, or recurrent neural networks when additional explanatory variables are available.

9 Conclusion

In this report, the weekly electricity consumption of the PJM East region was analyzed and forecasted by using an autoregressive integrated moving average-based time series analysis technique. The initial dataset comprises hourly electricity load records provided by the `PJME_hourly.csv` file. The dataset has been processed with respect to the time index, missing hourly readings, and incompleteness of some of the weeks. As a result, a weekly series has been obtained by summing up hourly load records.

The last version of the dataset has been divided into the training, validation, and testing sets in chronological order. This allows selecting the best model using the validation set while the test set will remain unseen until the final evaluation of forecast accuracy. Simple Naive and Seasonal Naive benchmarks have been applied to test whether the ARIMA model can significantly increase the forecast accuracy beyond trivial forecasting rules.

The ADF test demonstrates the stationarity of the weekly training series, which implies the use of the differencing order of $d = 0$. The optimal combination of parameters has been selected considering the performance of the validation set, model simplicity, and diagnostics. As a result, the final chosen model is ARIMA(3,0,2). The test forecast produced by this model has obtained MAE 250,006.38 MWh, RMSE 332,071.75 MWh, and MAPE 4.86% or the average forecast accuracy 95.14%. Moreover, the selected ARIMA(3,0,2) model is superior to the Naive and Seasonal Naive baselines according to the test set.

The one-step forecast performed by applying the rolling method has been considered as a supplementary forecasting technique aimed at detecting possible shortcomings of the selected multi-step model. The one-step rolling origin forecast obtained MAPE 4.45%. Thus, it should be concluded that the rolling origin forecasting approach based on ARIMA(3,0,2) model performs well enough. Nevertheless, multi-step forecasting remains a primary one due to its nature – forecasting of all test period points without incorporating real test data.

Thus, it is necessary to summarize that ARIMA(3,0,2) model is capable of producing accurate forecasts of weekly electricity consumption for the selected region. There are several limitations of the current model. First of all, it is purely univariate since no external factors such as temperature or weather conditions are considered. Furthermore, the ACF plot and residual diagnostics suggest the presence of annual seasonality around lag 52, which may not be fully captured by the selected non-seasonal ARIMA model.

Future analysis could extend this report by using SARIMAX or ARIMAX models with external variables, especially weather and holiday indicators. A longer rolling-origin evaluation and comparisons with machine learning models could also provide a more complete understanding of electricity consumption forecasting performance. Despite these limitations, the selected weekly ARIMA model provides strong out-of-sample forecasting accuracy and performs better than the simple baseline methods considered in this report.

References

Chodakowska, E., Nazarko, J., and Nazarko, L. (2021). Arima models in electrical load forecasting and their robustness to noise. *Energies*, 14(23):7952.

Elamin, N. and Fukushige, M. (2018). Modeling and forecasting hourly electricity demand by sarimax with interactions. *Energy*, 165:257–268.

Hyndman, R. J. and Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*. OTexts, 3 edition.

Izudin, N. E. M. et al. (2021). Forecasting electricity consumption in malaysia by hybrid arima-ann. In *Proceedings of the 6th International Conference on Fundamental and Applied Sciences*. Springer.

Lee, J. et al. (2021). National-scale electricity peak load forecasting. *arXiv preprint arXiv:2107.06174*.

Mulla, R. (2018). Hourly energy consumption. Kaggle Dataset. Accessed for PJME hourly electricity load data.

Tarmanini, C., Sarma, N., Gezegin, C., and Ozgonenel, O. (2023). Short term load forecasting based on arima and ann approaches. *Energy Reports*, 9:550–557.